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Dynamics modeling and motion control of 4IWMD/4WS vehicles based on torque coordination

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***Abstract*— This paper presents a dynamics model of electric vehicles with four in-wheel motor drive (4IWMD) and four-wheel steering (4WS), where the suspension geometry is incorporated. To explore the potential of utilizing the vertical reaction forces generated by the in-wheel motors' driving torques to control the vertical, pitch and roll motions of the vehicle body, torque coordination strategies are analyzed for both straight-line driving and cornering maneuver. A generalized torque distribution strategy is then derived to effectively and independently control each of the three body motions. Additionally, the influence of the suspension geometry on the maximum attainable vertical force, pitch moment and roll moment is investigated. The results can serve as the foundation for designing suspension geometry and motion control strategies for 4IWMD/4WS vehicles.**

I. Introduction

With the rapid development of motor technology, the innovative electrical propulsion system of in-wheel motor drive shows significant potential for vehicle electrification and intelligence [1]. Four in-wheel motor drive (4IWMD) vehicles can be easily integrated with four-wheel steering (4WS) systems, which facilitates improved handling performance [2]. Moreover, since the drive torque of each motor can be controlled independently, quickly and precisely, it is convenient to realize torque vectoring control to enhance the vehicle dynamics performance [3].

Due to the suspension geometry, the driving force of each wheel generates a vertical reaction force that acts on the vehicle body, thereby presenting the potential for controlling vertical, pitch and roll motions through coordinating the torque of each motor [4], [5]. Notably, this method does not require additional actuators, such as active suspension systems, making it a promising solution. Therefore, it is necessary to investigate the mechanism of vertical reaction forces and develop effective torque coordination strategies.

To reveal the effect of vertical reaction force on vehicle dynamics behavior, several models of 4IWMD vehicles have been proposed. A control-oriented model, which includes vertical, pitch and roll dynamics in response to the longitudinal and lateral tire forces, was developed in [6]. The controllability analysis and open-loop simulations were performed to demonstrate that the controlled driving forces can effectively influence the body motions. In [7] and [8], a roll dynamics model was built and a rollover prevention controller based on torque coordination was designed to enhance the lateral stability and handling. [9] modeled the relationship between the driving force and the targeted control forces or moments on the vehicle and proposed a decoupling method wherein the yaw, roll, and pitch or vertical motions are controlled independently using the reaction force of the suspension. In [10], [11], a quarter car model was developed and a skyhook damper control strategy based on torque coordination was designed to suppress the vertical vibration.

From the literature, it can be noted that previous research primarily focuses on developing dynamics models that consider suspension geometry for specific tasks and designing the corresponding motion control strategies. Few studies investigate the controllability conditions required to independently control each motion of the vehicle through torque coordination. Additionally, there has been limited research on the influence of suspension geometry on torque coordination strategies. Based on such considerations, this paper aims to develop a generalized dynamics model for 4IWMD/4WS vehicles that incorporates suspension geometry. The controllability conditions of the vehicle motions are then analyzed, and the corresponding torque coordination strategies are proposed. Finally, the effect of suspension geometry on attainable vertical force, pitch moment and roll moment is examined.

The remaining parts of this paper are organized as follows. Section II presents the modeling of suspension geometry and vehicle dynamics. Section III introduces the torque coordination strategy. Section IV investigates the effect of suspension geometry on the attainable forces and moments. Finally, Section V concludes the paper.

II. Modeling

In this section, the suspension geometry is described to elucidate the mechanism by which vertical reaction forces are generated in response to longitudinal and lateral forces. Then, a vehicle dynamics model is developed that incorporates these vertical reaction forces.

A. Mechanism of Vertical Reaction Forces

When a vehicle is accelerating, decelerating or cornering, the resulting inertial forces are causing lift, dive and roll motions of the body, which negatively impacting ride comfort and safety. To mitigate these effects, the suspension geometry is often adjusted so that the longitudinal and lateral forces generate a vertical component at the body, thereby reducing these motions [12]. The suspension geometry is depicted in Fig. 1. As noted by [13], any suspension can be functionally equivalent to a swing arm in the pitch-plane and roll-plane.

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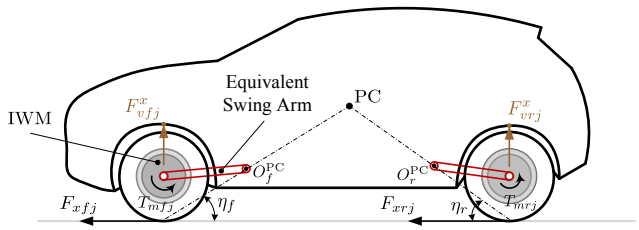
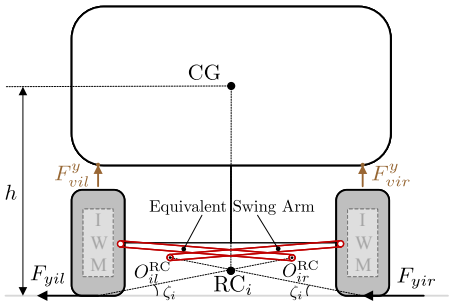


(a) Pitch-plane (side view)



(b) Roll-plane (rear view)

Fig. 1. Mechanism of vertical reaction forces

In the figure, O_i^{PC} and O_{ij}^{RC} represent the instantaneous centers of the suspension in the pitch-plane and roll-plane, respectively. CG, PC and RC denote the center of gravity, pitch center and roll center. The subscript $i \in \{f, r\}$ indicates the front/rear axle,and $j \in \{l, r\}$ denotes the left/right side. This notation will be used consistently throughout the paper.

The suspension geometry can be described by the angles between the ground and the lines connecting the tire grounding points to the instantaneous centers. These angles are denoted as η_i and ζ_i. Based on this definition, the vertical reaction force acting on the vehicle body can be derived as:

$$\begin{cases} F^x_{vfj} = -F_{xfj}\tan\eta_f \\ F^x_{vrj} = F_{xrj}\tan\eta_r \end{cases}, \begin{cases} F^y_{vil} = -F_{yil}\tan\zeta_i \\ F^y_{vir} = F_{yir}\tan\zeta_i \end{cases} \tag{1}$$

where F^x_{vij} is the vertical reaction force generated by the longitudinal force F_{xij}, and F^y_{vij} represents the vertical reaction force generated by the lateral force F_{yij}.

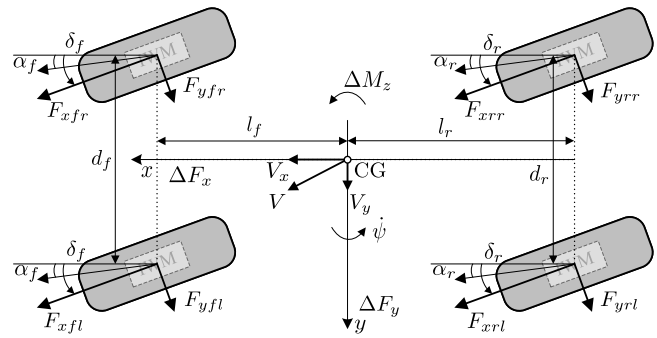
Consequently, the total vertical reaction force acting on the vehicle body can be expressed as:

$$F_{vij} = F^x_{vij} + F^y_{vij} \tag{2}$$

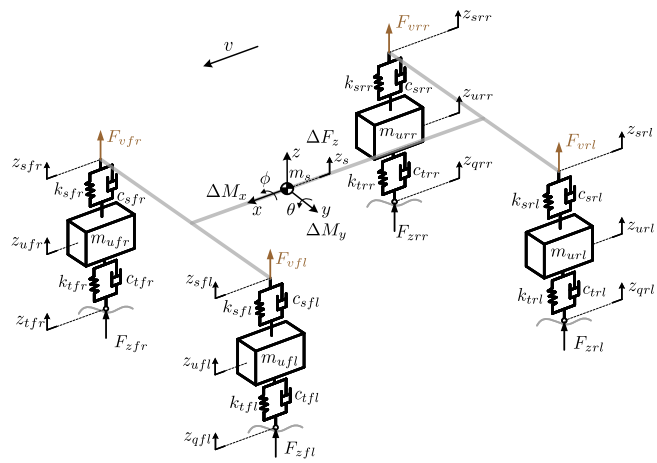
If the longitudinal and lateral forces at each wheel can be controlled, these forces can be utilized to effectively control the body motions, including vertical, pitch and roll motions. Specifically, in IWM-driven vehicles, the longitudinal force generates a larger vertical reaction force compared to conventional half-shaft-driven vehicles [1]. This significantly enhances the potential of IWM drive to control vehicle body motions.

B. Vehicle Dynamics Model with Vertical Reaction Forces

A 4IWMD/4WS vehicle dynamics model is developed, as shown in Fig. 2. The model in Fig. 2(a) primarily considers the longitudinal, lateral and yaw motions, while Fig. 2(b) focuses on the vertical motions of body and wheel masses.



(a) Longitudinal, lateral and yaw dynamics



(b) Vertical dynamics

Fig. 2. 4IWMD/4WS vehicle dynamics model

Assuming the left and right wheels on both the front and rear axles have the same steering angles, the longitudinal, lateral and yaw motions can be described as:

$$\begin{aligned} m(\dot{V}_x - V_y\dot{\psi}) = \sum\nolimits_j (&F_{xfj}\cos\delta_f - F_{yfj}\sin\delta_f \\ &+ F_{xrj}\cos\delta_r - F_{yrj}\sin\delta_r) + \Delta F_x \end{aligned} \tag{3}$$

$$\begin{aligned} m(\dot{V}_y + V_x\dot{\psi}) = \sum\nolimits_j (&F_{xfj}\sin\delta_f + F_{yfj}\cos\delta_f \\ &+ F_{xrj}\sin\delta_r + F_{yrj}\cos\delta_r) + \Delta F_y \end{aligned} \tag{4}$$

$$\begin{aligned} I_z\ddot{\psi} = \sum\nolimits_j &[l_f(F_{xfj}\sin\delta_f + F_{yfj}\cos\delta_f) - l_r(F_{xrj}\sin\delta_r \\ &+ F_{yrj}\cos\delta_r)] + \sum\nolimits_i \frac{d_i}{2}[(F_{xir} - F_{xil})\cos\delta_i \\ &+ (F_{yil} - F_{yir})\sin\delta_i] + \Delta M_z \end{aligned} \tag{5}$$

where ΔF_x, ΔF_y and ΔM_z represent the additional longitudinal force, lateral force and yaw moment generated by the IWMs and can be defined as:

$$\Delta F_x = (\Delta F_{xfl} + \Delta F_{xfr})\cos\delta_f + (\Delta F_{xrl} + \Delta F_{xrr})\cos\delta_r \tag{6}$$

$$\Delta F_y = (\Delta F_{xfl} + \Delta F_{xfr})\sin\delta_f + (\Delta F_{xrl} + \Delta F_{xrr})\sin\delta_r \tag{7}$$

$$\begin{aligned} \Delta M_z = &\sum\nolimits_i (\Delta F_{xir} - \Delta F_{xil})\frac{d_i}{2}\cos\delta_i \\ &+ \sum\nolimits_j (\Delta F_{xfj} l_f \sin\delta_f - \Delta F_{xrj} l_r \sin\delta_r) \end{aligned} \tag{8}$$

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The rotational motion of each wheel is given by:

$$J_w\dot{\omega}_{ij} = (T_{mij} + \Delta T_{mij}) - T_{bij} - (F_{xij} + \Delta F_{xij})R_w \quad (9)$$

where J_w is the moment of inertia of the wheel, ω_{ij} is the angular velocity of the wheel, T_{mij} is the motor drive torque, T_{bij} is the brake torque, and R_w is the wheel radius. ΔT_{mij} is the additional motor drive torque, which can be given by:

$$\Delta T_{mij} = \Delta F_{xij} R_w \quad (10)$$

With low lateral acceleration, the tires operate within the linear region. The lateral forces of the tires corresponding to the front and rear wheels can be expressed as:

$$F_{yij} = C_{\alpha i}\alpha_i \quad (11)$$

where

$$\alpha_f = \frac{V_y + l_f\dot{\psi}}{V_x} - \delta_f, \quad \alpha_r = \frac{V_y - l_r\dot{\psi}}{V_x} - \delta_r \quad (12)$$

For the vehicle body, referred to as the sprung mass, the vertical, pitch and roll dynamics are derived as:

$$m_s\ddot{z}_s = -\sum\nolimits_{i,j} F_{sij} + \sum\nolimits_{i,j} F_{vij} + \Delta F_z \quad (13)$$

$$\begin{aligned} I_y\ddot{\theta} = &\sum\nolimits_j (l_f F_{sfj} - l_r F_{srj} - l_f F_{vfj} + l_r F_{vrj}) \\ &- \sum\nolimits_{i,j} F_{xij} h + \Delta M_y \end{aligned} \quad (14)$$

$$\begin{aligned} I_x\ddot{\phi} = &\sum\nolimits_i \frac{d_i}{2}(-F_{sil} + F_{sir} + F_{vil} - F_{vir}) \\ &+ \sum\nolimits_{i,j} F_{yij} h + \Delta M_x \end{aligned} \quad (15)$$

where $F_{sij} = k_{sij}(z_{sij} - z_{uij}) + c_{sij}(\dot{z}_{sij} - \dot{z}_{uij})$ is the suspension force, and

$$\begin{cases} z_{sfl} = z_s - l_f\theta + \dfrac{d_f}{2}\phi, z_{sfr} = z_s - l_f\theta - \dfrac{d_f}{2}\phi \\ z_{srl} = z_s + l_r\theta + \dfrac{d_r}{2}\phi, z_{srr} = z_s + l_r\theta - \dfrac{d_r}{2}\phi \end{cases} \quad (16)$$

The additional vertical force ΔF_z, pitch moment ΔM_y and roll moment ΔM_z can be derived as:

$$\begin{aligned} \Delta F_z = &\sum\nolimits_j (\Delta F_{xrj}\tan\eta_r\cos\delta_r - \Delta F_{xfj}\tan\eta_f\cos\delta_f) \\ &+ \sum\nolimits_j (\Delta F_{xir} - \Delta F_{xil})\tan\zeta_i\sin\delta_i \end{aligned} \quad (17)$$

$$\begin{aligned} \Delta M_y = &\sum\nolimits_i (\Delta F_{xil} + \Delta F_{xir})(l_i\tan\eta_i - h)\cos\delta_i \\ &- (\Delta F_{xfr} - \Delta F_{xfl})l_f\tan\zeta_f\sin\delta_f \\ &+ (\Delta F_{xrr} - \Delta F_{xrl})l_r\tan\zeta_r\sin\delta_r \end{aligned} \quad (18)$$

$$\begin{aligned} \Delta M_z = &(\Delta F_{xfr} - \Delta F_{xfl})\frac{d_f}{2}\tan\eta_f\cos\delta_f \\ &- (\Delta F_{xrr} - \Delta F_{xrl})\frac{d_r}{2}\tan\eta_r\cos\delta_r \\ &- \sum\nolimits_i (\Delta F_{xil} + \Delta F_{xir})\left(\frac{d_i}{2}\tan\zeta_i - h\right)\sin\delta_i \end{aligned} \quad (19)$$

The vertical motion of the wheel can be described as:

$$m_{uij}\ddot{z}_{uij} = F_{sij} - F_{vij} - F_{tij} \quad (20)$$

where $F_{tij} = k_{tij}(z_{uij} - z_{qij}) + c_{tij}(\dot{z}_{uij} - \dot{z}_{qij})$ is the tire vertical force.

The vertical load of each wheel can be calculated as:

$$\begin{cases} F_{zfl} = \dfrac{mgl_r}{2L} - \dfrac{m\dot{V}_x h}{2L} - \dfrac{m\dot{V}_y h}{d_f}\dfrac{l_r}{L} + F_{vfl} \\ F_{zfr} = \dfrac{mgl_r}{2L} - \dfrac{m\dot{V}_x h}{2L} + \dfrac{m\dot{V}_y h}{d_f}\dfrac{l_r}{L} + F_{vfr} \\ F_{zrl} = \dfrac{mgl_f}{2L} + \dfrac{m\dot{V}_x h}{2L} - \dfrac{m\dot{V}_y h}{d_r}\dfrac{l_f}{L} + F_{vrl} \\ F_{zrr} = \dfrac{mgl_f}{2L} + \dfrac{m\dot{V}_x h}{2L} + \dfrac{m\dot{V}_y h}{d_r}\dfrac{l_f}{L} + F_{vrr} \end{cases} \quad (21)$$

where $L = l_f + l_r$ is the wheelbase. The symbols used in the modeling and their definitions are listed in Table I.

TABLE I
SYMBOLS DEFINITION

Symbols	Definition/Unit
m, m_s	Vehicle mass, sprung mass (kg)
I_x, I_y, I_z	Yaw/Pitch/Roll moment of inertia of vehicle (kg·m^2)
ψ, θ, ϕ	Yaw/Pitch/Roll angle (rad)
δ_f, δ_r	Steering angles of front and rear wheels (rad)
α_f, α_r	Sideslip angles of front and rear wheels (rad)
l_f, l_r	Distances from the front and rear axles to the CG (m)
d_f, d_r	Distances between front/rear left and right wheels (m)
h	Distance between CG and ground (m)
$C_{\alpha i}$	Cornering stiffness of the tire (N/rad)
k_{sij}, k_{tij}	Stiffness of vehicle suspension and tire (N/m)
c_{sij}, c_{tij}	Damping coefficient of vehicle suspension and tire (N·s/m)
z_s, z_{uij}	Vertical displacement of sprung mass and wheel (m)
z_{qij}	Road disturbance input (m)

III. TORQUE COORDINATION STRATEGY

In this section, the torque coordination strategies for vehicle motion control during straight-line driving and cornering maneuver are analyzed. Then, the generalized torque distribution strategy is derived.

A. Straight-line Driving

During straight-line driving, the vehicle body is subject to vertical, roll and pitch motions induced by road excitations. Consequently, the control targets for straight-line driving include longitudinal, vertical, roll and pitch motions. The relationship between the control driving force and the targeted control forces or moments can be expressed as:

$$y_4^s = C_4^s u_x \quad (22)$$

where

$$y_4^s = [\Delta F_x, \Delta F_z, \Delta M_x, \Delta M_y]^{\mathrm{T}},$$

$$u_x = [\Delta F_{xfl}, \Delta F_{xfr}, \Delta F_{xrl}, \Delta F_{xrr}]^{\mathrm{T}},$$

$$C_4^s = \begin{bmatrix} 1 & 1 & 1 & 1 \\ -\eta_f^t & -\eta_f^t & \eta_r^t & \eta_r^t \\ -\dfrac{d_f}{2}\eta_f^t & \dfrac{d_f}{2}\eta_f^t & \dfrac{d_r}{2}\eta_r^t & -\dfrac{d_r}{2}\eta_r^t \\ l_f\eta_f^t - h & l_f\eta_f^t - h & l_r\eta_r^t - h & l_r\eta_r^t - h \end{bmatrix}$$

For simplicity, η_i^t denotes $\tan\eta_i$.

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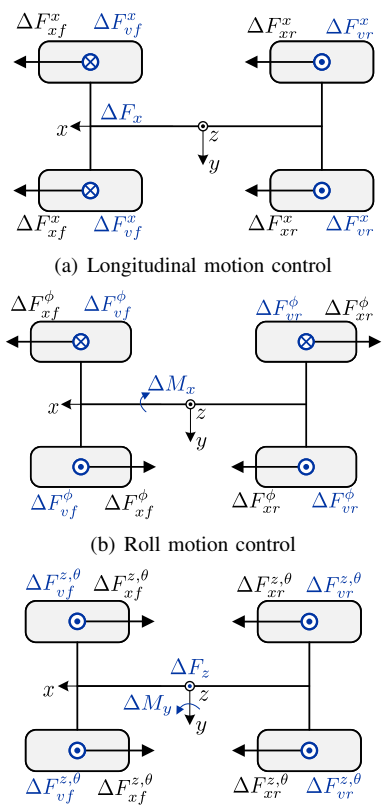
To independently control the vehicle's motions in these four directions by coordinating the driving torque of each wheel, the matrix C_4^s needs to be full rank. However, the rank of matrix C_4^s is 3. Let $C_4^s(i)$ denote the ith row of C_4^s, then we have:

$$\left(\frac{(l_f+l_r)\eta_f^t\eta_r^t}{\eta_f^t+\eta_r^t}-h\right)C_4^s(1)+\frac{l_r\eta_r^t-l_f\eta_f^t}{\eta_f^t+\eta_r^t}C_4^s(2)=C_4^s(4) \tag{23}$$

This indicates that the 1st, 2nd and 4th rows of C_4^s are linearly correlated. Assuming $\tan\eta_i>0$, it can be observed that $\mathrm{rank}(C_4^s(1,2,4))=2$, $\mathrm{rank}(C_4^s(1,2,3))=3$ and $\mathrm{rank}(C_4^s(1,3,4))=3$. This reveals that the vertical and pitch motions cannot be independently controlled by torque coordination. Therefore, one of the pitch or vertical motions can be selected, along with the longitudinal and roll motions, to form the control targets during straight-line driving. The selection of control targets is also influenced by the required dynamic performance and the vehicle parameters. For example, when $l_f\eta_f^t=l_r\eta_r^t$, it becomes impossible to control the pitch motion as $\mathrm{rank}(C_4^s(1,4))=1$.

Then, the decoupled torque coordination strategy for straight-line driving can be derived based on C_4^s, as shown in Fig. 3.



(a) Longitudinal motion control

(b) Roll motion control

(c) Vertical/Pitch motion control

Fig. 3. Torque coordination strategy for straight-line driving

It can be noted that when controlling the roll motion and vertical/pitch motion, if $|\Delta F_{xf}^\phi|d_f=|\Delta F_{xr}^\phi|d_r$ and $|\Delta F_{xf}^{z,\theta}|=|\Delta F_{xr}^{z,\theta}|$, the control forces in the longitudinal direction cancel out, as do the moments generated by the front and rear axles. Therefore, they have no additional effect on the longitudinal and yaw motions. However, torque coordination strategy in Fig. 3(c) will influence both vertical and pitch motions, which requires careful considerations in the controller design.

B. Cornering Maneuver

Compared to straight-line driving, cornering maneuver also induces yaw and lateral motions. Due to the inefficiency of directly controlling lateral motion using motor drive torque and steering, lateral motion is excluded from the control targets. Thus, the primary control targets during cornering include longitudinal, yaw, roll, vertical and pitch motions. The control inputs consist of the front and rear steering angles as well as the motor drive torque. Since the steering input indicates the intended driving direction, it is treated as a fixed value in the following analysis rather than a control input. The relationship between the control driving force and the targeted control forces or moments can be derived as:

$$y_5^c=C_5^c u_x \tag{24}$$

where $y_5^c=[\Delta F_x,\Delta F_z,\Delta M_x,\Delta M_y,\Delta M_z]^{\mathrm{T}}$,

$$C_5^c=\begin{bmatrix} \delta_f^c & \delta_f^c & \delta_r^c & \delta_r^c \\ -\eta_f^t\delta_f^c-\zeta_f^t\delta_f^s & -\eta_f^t\delta_f^c+\zeta_f^t\delta_f^s & \eta_r^t\delta_r^c-\zeta_r^t\delta_r^s & \eta_r^t\delta_r^c+\zeta_r^t\delta_r^s \\ -\frac{d_f}{2}\eta_f^t\delta_f^c-\left(\frac{d_f}{2}\zeta_f^t-h\right)\delta_f^s & \frac{d_f}{2}\eta_f^t\delta_f^c-\left(\frac{d_f}{2}\zeta_f^t-h\right)\delta_f^s & \frac{d_r}{2}\eta_r^t\delta_r^c-\left(\frac{d_r}{2}\zeta_r^t-h\right)\delta_r^s & -\frac{d_r}{2}\eta_r^t\delta_r^c-\left(\frac{d_r}{2}\zeta_r^t-h\right)\delta_r^s \\ \left(l_f\eta_f^t-h\right)\delta_f^c+l_f\zeta_f^t\delta_f^s & \left(l_f\eta_f^t-h\right)\delta_f^c-l_f\zeta_f^t\delta_f^s & (l_r\eta_r^t-h)\delta_r^c-l_r\zeta_r^t\delta_r^s & (l_r\eta_r^t-h)\delta_r^c+l_r\zeta_r^t\delta_r^s \\ l_f\delta_f^s-\frac{d_f}{2}\delta_f^c & l_f\delta_f^s+\frac{d_f}{2}\delta_f^c & l_r\delta_r^s-\frac{d_r}{2}\delta_r^c & l_r\delta_r^s+\frac{d_r}{2}\delta_r^c \end{bmatrix}$$

For simplicity, ζ_i^t, δ_i^c and δ_i^s denote $\tan\zeta_i$, $\cos\delta_i$ and $\sin\delta_i$, respectively.

It should be noted that there are four control inputs, thus only four target motions can be controlled. For handling and safety considerations, the primary targets are longitudinal, yaw and roll motions. The rank of $C_5^c(1,3,5)$ is 3, indicating that these three motions can be controlled independently.

Then, one of the vertical and pitch motions can be selected as the remaining control target. Since $\mathrm{rank}(C_5^c(1,3,4,5))=4$ and $\mathrm{rank}(C_5^c(1,2,3,5))=4$, it is demonstrated that independent control can be achieved by incorporating either vertical or pitch motion into the control targets. However, if the vehicle's front/rear suspension parameters and the weight distribution are symmetrical, and $\zeta_i=0$ or $l_i\eta_i^t=h$, the rank of $C_5^c(1,3,4,5)$ is reduced to 3. This indicates that independent control of the pitch motion is not possible in this case. Therefore, whether independent control of the selected targets is achievable depends on the specific vehicle parameters and requires further analysis.

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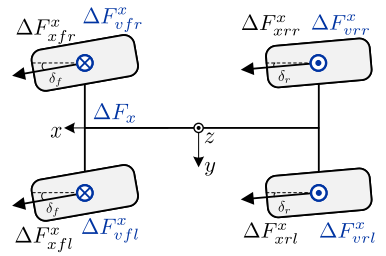
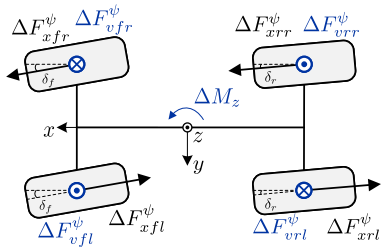
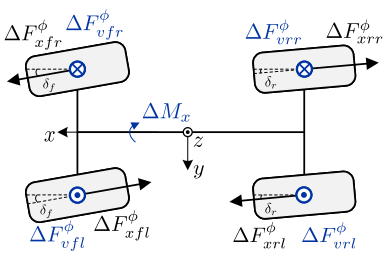
The decoupled torque coordination strategy for cornering maneuver is illustrated in Fig. 4. In the figure, the control forces for each motion are calculated through matrix C_5^c, based on vehicle parameters and steering angles.



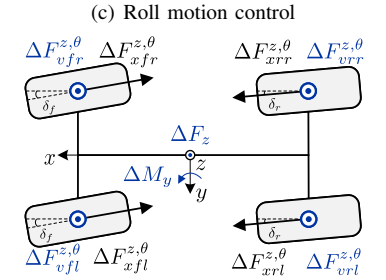
(a) Longitudinal motion control



(b) Yaw motion control



(c) Roll motion control



(d) Vertical/Pitch motion control

Fig. 4. Torque coordination strategy during cornering

C. Torque Distribution

Based on the torque coordination strategy for each motion, the distributed drive torque for each IWM can be derived as:

$$\begin{cases} \Delta T_{mfl} = (\Delta F^x_{xfl} - \Delta F^\psi_{xfl} - \Delta F^\phi_{xfl} - \Delta F^{z,\theta}_{xfl})R_w \\ \Delta T_{mfr} = (\Delta F^x_{xfr} + \Delta F^\psi_{xfr} + \Delta F^\phi_{xfr} - \Delta F^{z,\theta}_{xfr})R_w \\ \Delta T_{mrl} = (\Delta F^x_{xrl} - \Delta F^\psi_{xrl} + \Delta F^\phi_{xrl} + \Delta F^{z,\theta}_{xrl})R_w \\ \Delta T_{mrr} = (\Delta F^x_{xrr} + \Delta F^\psi_{xrr} - \Delta F^\phi_{xrr} + \Delta F^{z,\theta}_{xrr})R_w \end{cases} \tag{25}$$

Since the control torque for each motion is calculated independently, the motor torque or tire adhesion may exceed the limits. Therefore, in practical applications, it is essential to ensure that the control torque adheres to the following constraints:

$$\begin{cases} |T_{mij} + \Delta T_{mij}| \le T_{\max} \\ \sqrt{(F_{xij} + \Delta F_{xij})^2 + F^2_{yij}} \le \mu F_{zij} \end{cases} \tag{26}$$

where μ is the friction coefficient.

IV. Attainable Force/Moment Analysis

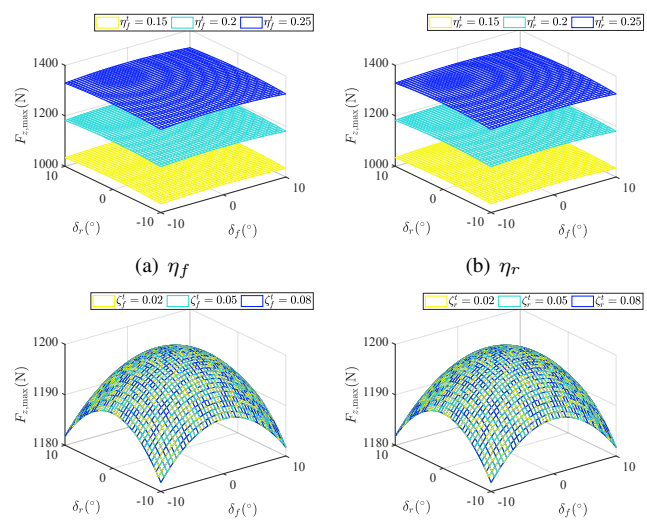
In this section, the influence of suspension geometry on the maximum attainable forces and moments is analyzed to explore the potential of vertical reaction force in controlling body motions. A set of vehicle parameters is listed in Table II, and the suspension geometry is varied to calculate the maximum attainable vertical force, roll moment and pitch moment. It is assumed that both the front and rear steering angles vary from -10° to 10°. The results are discussed in Section IV-A to IV-C.

TABLE II
Vehicle Parameters

Parameter	Value/Unit	Parameter	Value/Unit	Parameter	Value/Unit
l_f	1.2 m	η^t_f	0.2	h	0.35 m
l_r	1.7 m	η^t_r	0.2	R_w	0.33 m
d_f	1.6 m	ζ^t_f	0.05	$T_{\max}$	500 N·m
d_r	1.6 m	ζ^t_r	0.05		

A. Maximum Attainable Vertical Force

The maximum attainable vertical forces with different suspension geometries are shown in Fig. 5.



(a) η_f (b) η_r

(c) ζ_f (d) ζ_r

Fig. 5. Maximum attainable vertical forces

The figure illustrates that the maximum vertical force increases with η^t_i and reaches its peak when the front and rear steering angles are zero. However, the suspension geometry in the roll-plane has no effect on the maximum vertical force. This is because the lateral forces generated by the torque coordination strategy in Fig. 4(d) are in the same direction,

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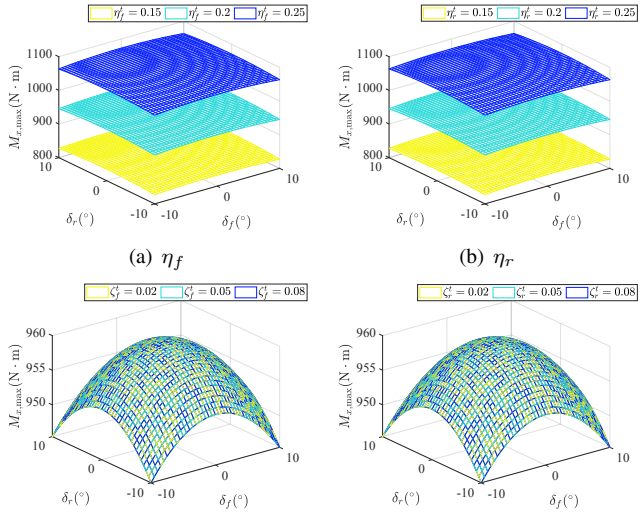
resulting in the cancellation of the vertical reaction forces within the roll-plane.

B. Maximum Attainable Roll Moment

The maximum roll moments with different suspension geometries are shown in Fig. 6.



(a) η_f (b) η_r

(c) ζ_f (d) ζ_r

Fig. 6. Maximum attainable roll moments

The results exhibit a similar pattern to the maximum attainable vertical forces, with the maximum attainable roll moment increases as η_i^t increases. Additionally, the suspension geometry in the roll-plane has no influence on the maximum roll moment. This occurs because the torque coordination for roll motion control in Fig. 4(c) does not produce additional roll moment within the roll-plane.

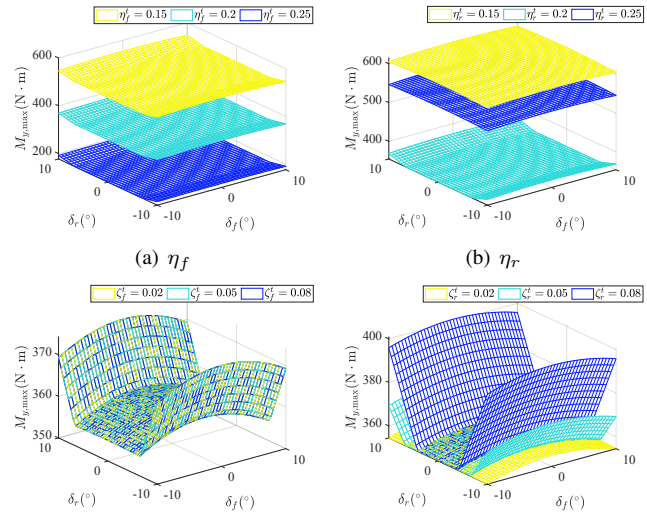
C. Maximum Attainable Pitch Moment

The maximum pitch moments with different suspension geometries are shown in Fig. 7. The figure indicates that the maximum attainable pitch moment decreases with an increase in η_f^t, but initially decreases and then increases with an increase in η_r^t. Moreover, ζ_r^t has no effect on the maximum attainable pitch moment, whereas an increase in ζ_r^t leads to an increasing trend in the maximum attainable pitch moment at large steering angles. This is because the given vehicle parameters cause the coefficients of the pitch moment in C_5^c to cross zero, resulting in an inflection point in the maximum attainable pitch moment.

The above analysis indicates that the behavior of maximum attainable forces and moments requires specific analysis in conjunction with detailed vehicle parameters.

V. Conclusions

This paper develops a dynamics model for a 4IWMD/4WS vehicle that incorporates suspension geometry. It examines controllability and torque coordination strategies for both straight-line and cornering maneuvers, and investigates the impact of suspension geometry on the maximum attainable forces and moments. The results provide valuable insights for designing suspensions and motion control strategies.



(a) η_f (b) η_r

(c) ζ_f (d) ζ_r

Fig. 7. Maximum attainable pitch moments

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